Hasan Haider

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Work Experience

GeoAnalysis Engineering - FrontEnd React Developer — Jul 2023 – Jun 2024

React, Tailwind CSS, JavaScript, Flowbite, Node.js

- Developed a fully responsive website from client requirements using modern frontend technologies.
- Gathered requirements, refactored the legacy codebase, and built a parallel site that replaced the original.
- Improved performance and loading time by **55%** while optimizing layouts for multiple screen sizes.

GAE Internship - Software Engineering Intern — May 2025 – Aug 2025

Python, PyInstaller, Git, Debugging

- Refactored the Intelligent Thermal Conductivity Meter from the ground up, improving modularity and reducing loading time by **38%**.
- Reduced active threads by **25%** through cleaner threading logic and optimized runtime behavior.
- Built a lightweight, thread-safe live data plotter for real-time instrument readings.
- Created automated error handling and a PyInstaller pipeline for generating easy-to-use Windows installers.

Projects

Earth Orbit Live – Satellite & Starlink Tracker — Jul 2026

React, TypeScript, Three.js, satellite.js, WebGL, Cloudflare Workers

- Built a responsive 3D visualization of thousands of Earth-orbiting satellites with interactive constellation filtering.
- Propagated live CelesTrak TLE data in the browser using SGP4 and synchronized satellite positions with simulation time.
- Created a cached Cloudflare Worker proxy and clearly labeled simulated fallbacks for unavailable data sources.
- Added pause, reverse, date selection, zoom, and logarithmic speed controls with real-time status and performance telemetry.

Motion Labs — Jun 2026

React, Three.js, Node.js, WebSocket, SSE, DeviceMotion API

- Created a motion lab pairing a phone with a desktop to turn live sensor data into a 3D sword.
- Built QR-based phone pairing with HTTPS-aware routing, session IDs, and hosted/local relay fallback.
- Streamed DeviceMotion and DeviceOrientation data through WebSocket/SSE with telemetry and reconnect handling.
- Mapped calibrated phone motion into a Three.js sword scene with hit detection, scoring, and live readouts.

Nearby Bluetooth – BLE Scanner & Signal Finder — Aug 2026

React Native, TypeScript, react-native-ble-plx, Android, Bluetooth Low Energy

- Built a React Native Android app that discovers nearby BLE advertisers with runtime-permission and adapter-state handling.
- Preserved first-seen device order during timed discovery and added manual rescans to prevent RSSI-based list movement.
- Added focused tracking with smoothed live RSSI, numeric dBm output, a proximity meter, and stale-signal detection.
- Bundled a standalone release APK for on-device use without USB or the Metro development server.

Ollama-Based Multimedia Edubot — Jan 2026 – Apr 2026

Python, Wikimedia API, ChromaDB, Ollama, SQL, LLMs, Taiga

- Led a team of 6 to add multimedia capabilities to an educational chatbot by retrieving Wikimedia images instead of generating AI images.
- Built a ChromaDB image cache for faster retrieval and reduced operational costs.
- Added offline system support using Ollama and implemented an Online/Offline mode switch.
- Coordinated planning through Taiga boards, weekly scrum meetings, and agile workflows.

Education

M.S. Software Engineering, Cybersecurity Specialization — Aug 2024 – May 2026

Arizona State University, Tempe, AZ, USA GPA: 3.5/4

B.Tech Information Technology — Aug 2019 – Jul 2023

KIET Group of Institutions, Delhi NCR, India GPA: 7.32/10

Skills

Languages: Python, C/C++, C#, JavaScript, TypeScript, Swift, Kotlin, SQL, HTML/CSS
AI Tools: Claude Code, Cursor, Codex
Frontend: React, React Native, Tailwind CSS, Flowbite, WebGL
Backend & Tools: Node.js, PyInstaller, Selenium, Git, Arduino, ESP8266, Unity, FFmpeg, Linux, WebSocket
Other: OCR with Tesseract, DeepL API, Neural Networks, Real-Time Plotting, Thread Safety, LLMs, CUDA

Journal Publications

- Rizvi Z.H., Akhtar S.J., Husain S.M.B., Khan M., **Haider, H.**, Naqvi S., Tirth V., Wuttke F. (2022). Neural network approaches for computation of soil thermal conductivity. *Mathematics*, 10(21), 3957.
- Rizvi, Z.H., Baqir Husain, S.M., **Haider, H.**, Wuttke, F. (2020). Effective thermal conductivity of sands estimated by Group Method of Data Handling. *Materials Today: Proceedings*, 26(2), 2103–2107.
- Rizvi, Z.H., Akhtar, S.J., **Haider, H.**, Follmann, J., Wuttke, F. (2019). Estimation of seismic wave velocities of metamorphic rocks using artificial neural network. *Materials Today: Proceedings*, 26(2), 324–330.